How-To Library

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Performance Tuning Guidelines for PowerExchange for JDBC V2 on the Spark Engine

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Abstract

Use PowerExchange for JDBC V2 to read from and write to Aurora PostgreSQL, Azure SQL Database, and databases with the Type 4 JDBC driver. Multiple factors such as hardware parameters, database parameters, and mapping parameters impact the adapter performance. You can optimize the performance by tuning these parameters appropriately.

This article describes general reference guidelines to help you tune the performance of PowerExchange for JDBC V2 on the Spark engine.

Supported Versions

- Informatica® PowerExchange® for JDBC V2

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Overview

Performance tuning is an iterative process where you analyze the performance, use guidelines to estimate and define parameters that impact the performance, monitor, and adjust the results as required. This document describes the parameters that you can tune to optimize the performance of PowerExchange for JDBC V2 on the Spark engine.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of PowerExchange for JDBC V2 might vary based on individual environments and other parameters even when you use the same data.

Performance Tuning Areas

You can optimize the performance of PowerExchange for JDBC V2 mappings by tuning the following areas:

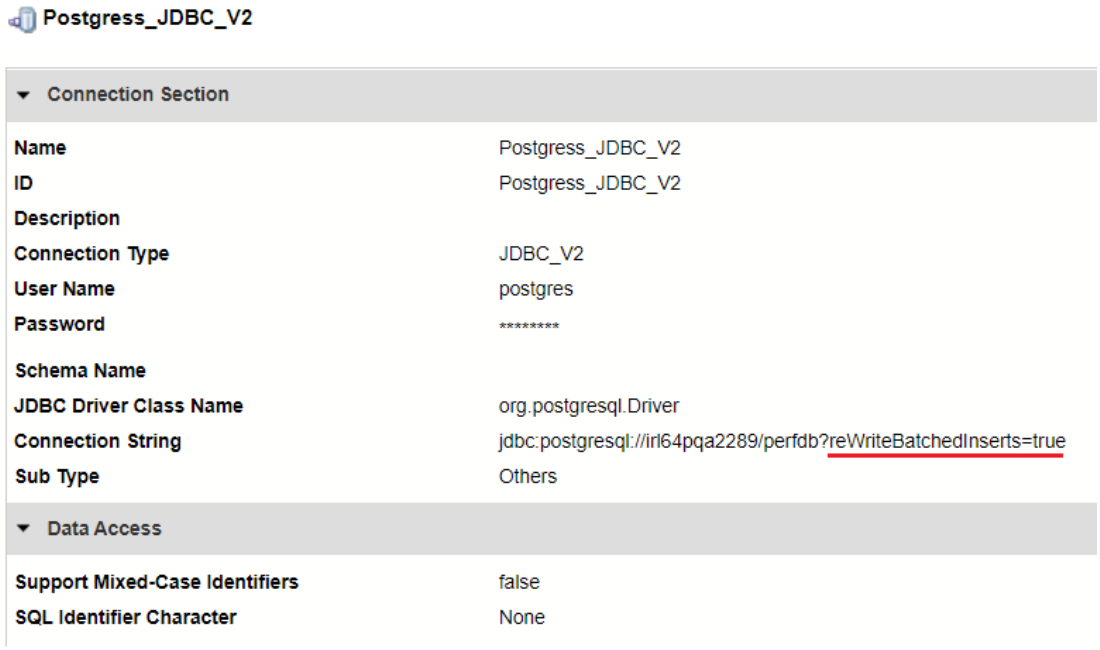
- Driver
- Mapping

Tune the Driver Parameters

You can configure the driver parameters to tune the performance of PowerExchange for JDBC V2.

For example, when you configure a JDBC V2 connection to connect to PostgreSQL, you can specify the reWriteBatchedInserts property in the connection string.

The following image shows the reWriteBatchedInserts property in the JDBC V2 connection string:



Tune the Mapping

You can configure the mapping parameters to tune the performance of PowerExchange for JDBC V2.

Partitioning

You can configure partitioning for the read or write operation in a JDBC V2 mapping to optimize the mapping performance. The partition type controls how the Spark Engine distributes data among partitions at partition points.

Partition Key

You can define a column as the partition key. To optimize performance, select a column that has diverse value as the partition key. You can configure a partition key only for fields of the numeric data type, such as int, decimal, or number.

Perform the following steps to define the partition key:

1. Select the **Overview** tab for the read or write operation.
2. Under **Object Details**, select the **Columns** tab.

3. Select a column that you want to assign as the partition key.
   The following image shows the partition key in the **Columns** tab:

Object Details:

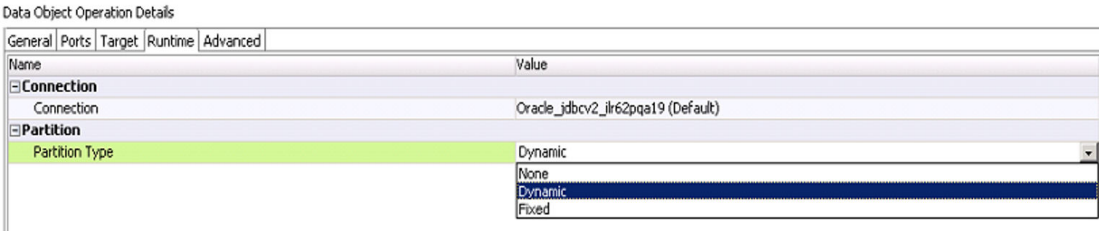
General | Columns

	Name	Native Name	Type	Precision	Scale	Partition Key	Access Type
1	L_ORDERKEY	L_ORDERKEY	numeric	38	0	☑	Read and Write
2	L_PARTKEY	L_PARTKEY	numeric	38	0	☐	Read and Write
3	L_SUPPKEY	L_SUPPKEY	numeric	38	0	☐	Read and Write
4	L_LINENUMBER	L_LINENUMBER	numeric	38	0	☐	Read and Write
5	L_QUANTITY	L_QUANTITY	numeric	10	2	☐	Read and Write
6	L_EXTENDEDPRICE	L_EXTENDEDP...	numeric	10	2	☐	Read and Write
7	L_DISCOUNT	L_DISCOUNT	numeric	10	2	☐	Read and Write
8	L_TAX	L_TAX	numeric	10	2	☐	Read and Write
9	L_RETURNFLAG	L_RETURNFLAG	varchar	1	0	☐	Read and Write
10	L_LINESTATUS	L_LINESTATUS	varchar	1	0	☐	Read and Write
11	L_SHIPDATE	L_SHIPDATE	varchar	29	0	☐	Read and Write
12	L_COMMITDATE	L_COMMITDATE	varchar	29	0	☐	Read and Write
13	L_RECEIPTDATE	L_RECEIPTDATE	varchar	29	0	☐	Read and Write
14	L_SHIPINSTRUCT	L_SHIPINSTRU...	varchar	25	0	☐	Read and Write
15	L_SHIPMODE	L_SHIPMODE	varchar	10	0	☐	Read and Write
16	L_COMMENT	L_COMMENT	varchar	44	0	☐	Read and Write

Partition Type

You can configure dynamic partitioning or fixed partitioning. To configure the partitioning, open the JDBC V2 data object read or write operation and on the **Runtime** tab, select **Fixed** or **Dynamic**.

The following image shows the partition types in the runtime properties:



Dynamic Partitioning

You can configure dynamic partitioning only for the write operation to optimize the mapping performance.

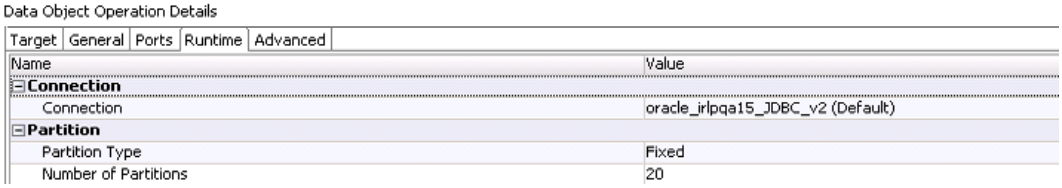
Note: When you select dynamic partitioning, the Spark Engine creates the number of partitions for the target based on the source partitioning.

Fixed Partitioning

To configure fixed partitioning, specify the number of partitions that the Spark Engine creates at runtime. Select a column that you want to define as the partition key.

To optimize performance, set the number of partitions based on the available resources on both the Hadoop cluster and the database, that reduces the input-output wait time.

The following image shows the fixed partition type in the runtime properties:



Note: When you write to a database, you can specify the number of partitions that the Spark Engine creates for the target. If you do not specify the number of partitions, by default, the value is two.

Guidelines for Partitioning

Consider the following guidelines for partitioning to optimize performance:

- The number of partitions must be equal to the number of VCores available on the Hadoop cluster and the number of parallel connections the database can handle.
- When you configure the read operation, ensure that the constraints are enabled in the source table.

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